

Mortality models

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De Moivre Law

- The **De Moivre Law** assumes that the cohort function is linear with respect to the age

$$l_x = l_0(1 - x/\omega)$$

where ω is the actuarial infinity and l_0 is the initial cohort size. Based on that assumption, the rest of the biometric functions can be obtained as:

- $d_x = l_x - l_{x+1} = l_0/\omega$
- $\mu_x = \frac{-l'_x}{l_x} = \frac{1}{\omega-x}$
- ${}_np_x = \frac{l_{x+n}}{l_x} = \frac{\omega-x-n}{\omega-x} = 1 - n \cdot \mu_x$
- ${}_nq_x = 1 - {}_np_x = n \cdot \mu_x$

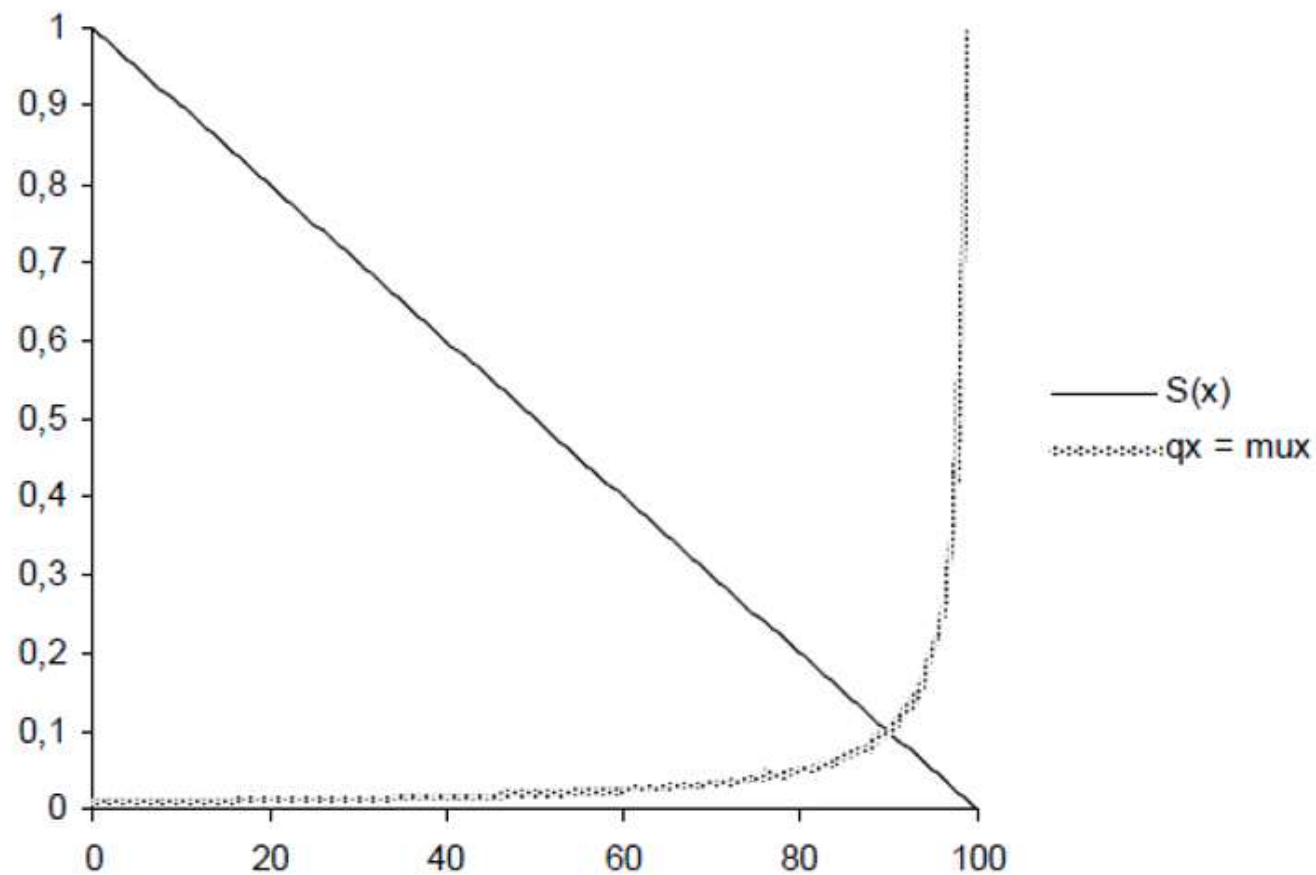
- $S_x = \frac{l_x}{l_0} = 1 - x/\omega$

- $f_x = \frac{1}{\omega}$

- $g_x(t) = {}_tp_x \cdot \mu_{x+t} = \frac{1}{\omega-x}$

- $\bar{e}(x) = \int_0^{\omega-x} t g_x(t) dt = \frac{\omega-x}{2}$

De Moivre Law (for $\omega = 100$)



First Law of Dormoy

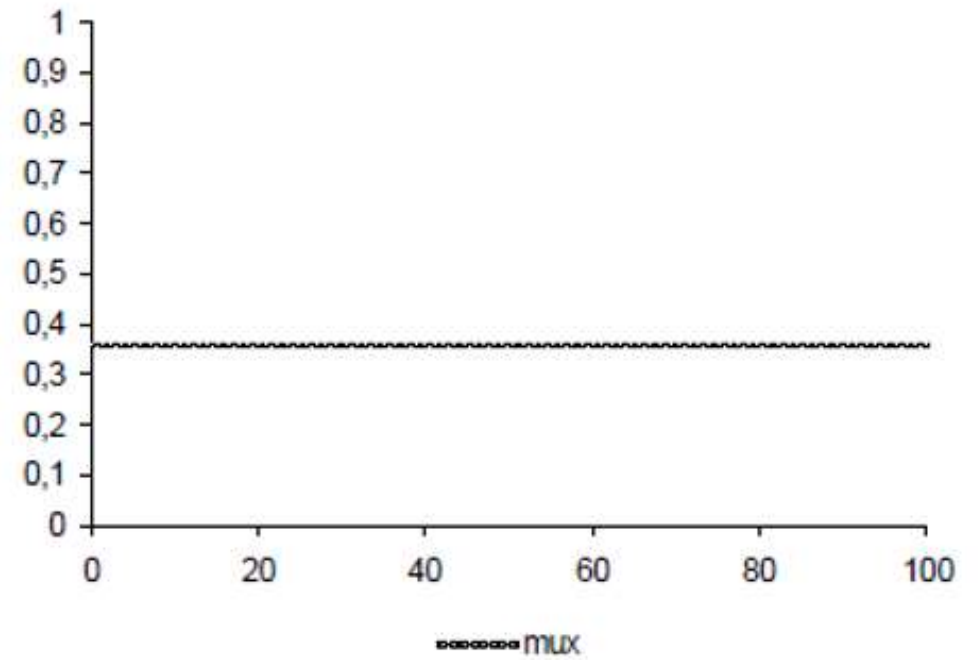
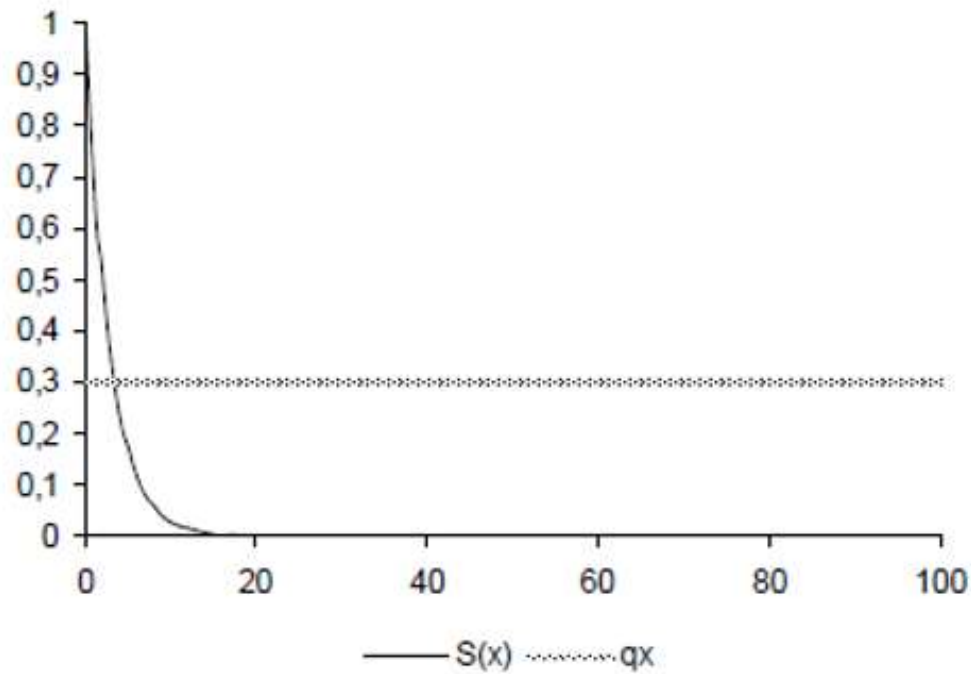
- The **First Law of Dormoy** assumes the following expression for the cohort function

$$l_x = l_0 S^x$$

where $S < 1$, so that l_x decreases with x . Based on that assumption, the rest of the biometric functions can be obtained as:

- $\mu_x = -\ln S$
- ${}_n p_x = S^n; {}_n q_x = 1 - S^n$
- $S(x) = S^x$
- $f(x) = -S^x \ln S$
- $g_x(t) = -S^t \ln S$
- $\bar{e}(x) = -1/\ln S$

First Law of Dormoy (for $S = 0.7$)



Second Law of Dormoy

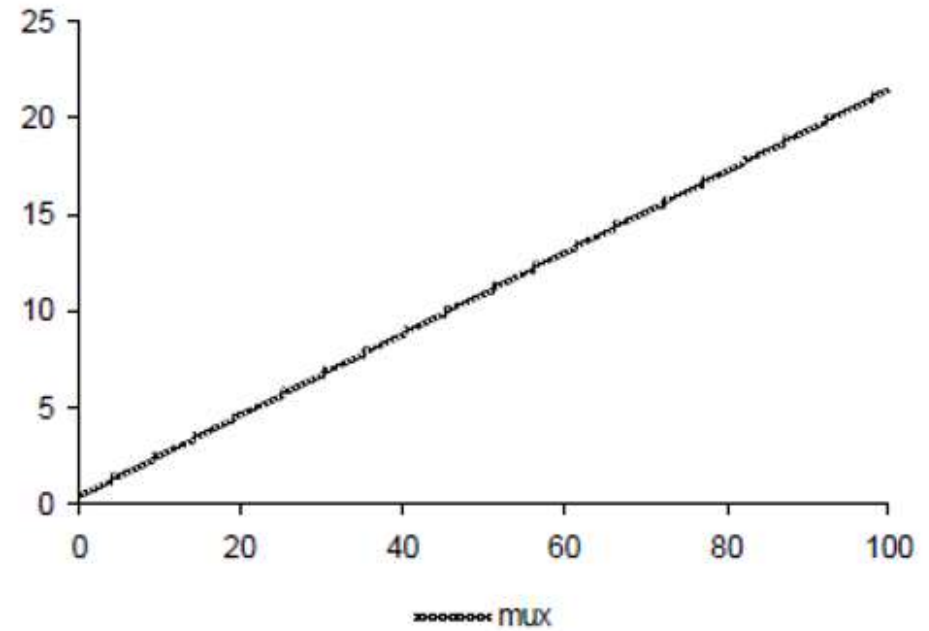
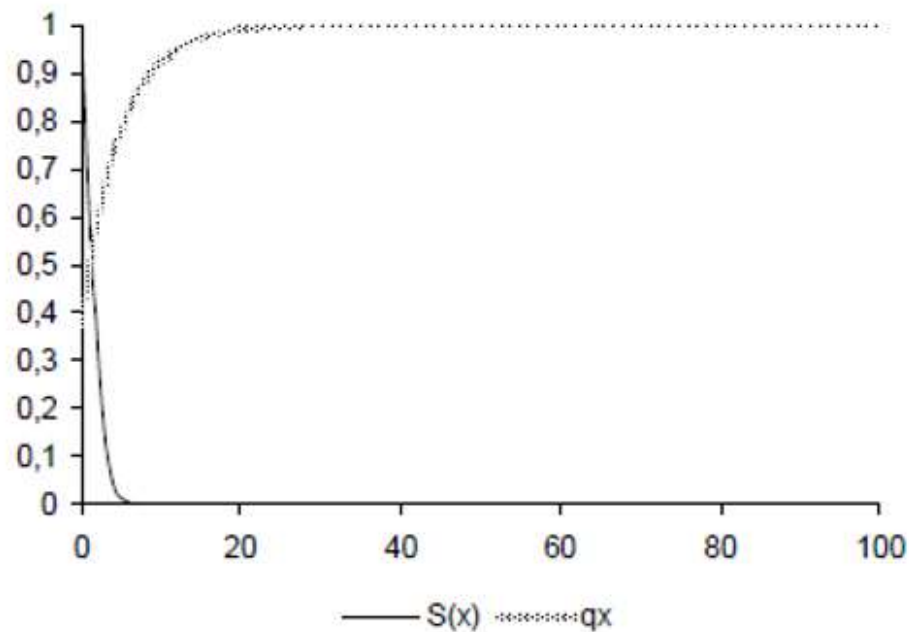
- The **Second Law of Dormoy** assumes the following expression for the cohort function

$$l_x = l_0 S_1^x S_2^{x^2}$$

where $S_1 < 1$ and $S_2 < 1$. Based on that assumption, the rest of the biometric functions can be obtained as:

- $\mu_x = -[2x \ln S_2 + \ln S_1]$
- ${}_n p_x = [S_1 S_2^{2x+n}]$; ${}_n q_x = 1 - [S_1 S_2^{2x+n}]$
- $S(x) = S_1^x S_2^{x^2}$
- $f(x) = -[2x \ln S_2 + \ln S_1] S_1^x S_2^{x^2}$
- $g_x(t) = -[2(x+t) \ln S_2 + \ln S_1] [S_1 S_2^{2x+t}]^t$
- $\bar{e}(x) = \int_0^\infty [S_1 S_2^{2x+t}]^t dt$

Second Law of Dormoy (for $S_1 = 0.7$; $S_2 = 0.9$)



Gompertz Law

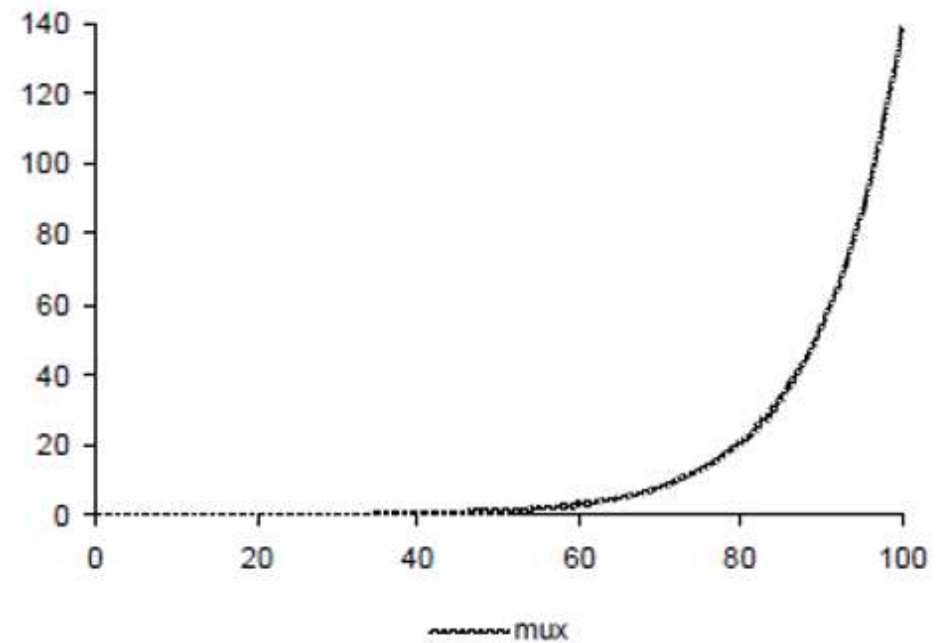
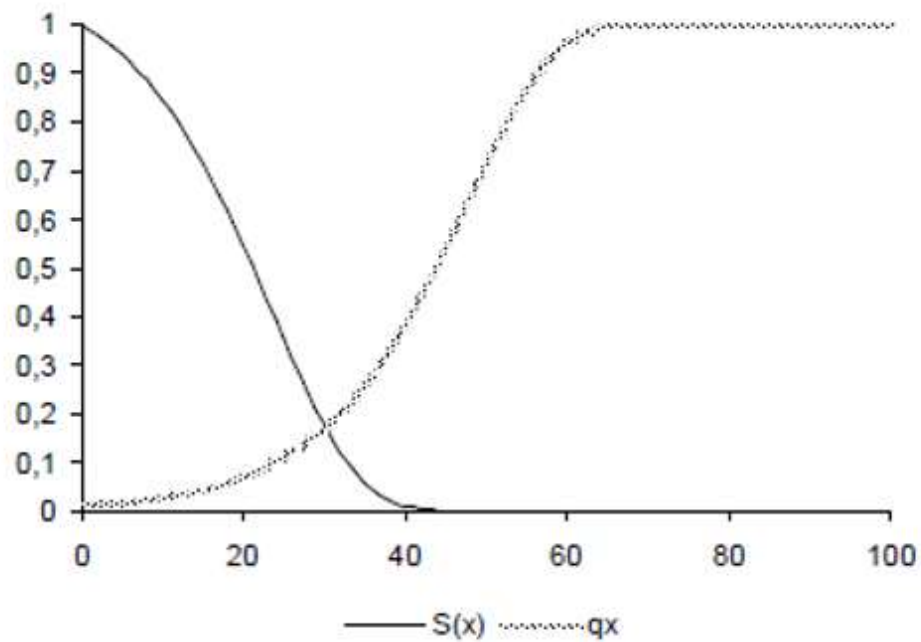
- The **Gompertz Law** assumes the following expression for the force of mortality

$$\mu_x = -\ln g \cdot \ln C \cdot C^x$$

where $g < 1$ and $C > 1$. The force of mortality increases with age, and its relative increase is constant. Based on that assumption, we have that:

- $l_x = l_0 \cdot g^{C^x - 1}$
- ${}_n p_x = g^{C^x(C^n - 1)}$; ${}_n q_x = 1 - g^{C^x(C^n - 1)}$
- $S(x) = g^{C^x - 1}$
- $f(x) = -\ln g \cdot \ln C \cdot C^x \cdot g^{C^x - 1}$
- $g_x(t) = -\ln g \cdot \ln C \cdot C^{x+t} \cdot g^{C^x(C^t - 1)}$
- $\bar{e}(x) = \int_0^\infty g^{C^x(C^t - 1)} dt$

Gompertz Law (for $g = 0.9$; $C = 1.1$)



Makeham Law

- The **Makeham Law** assumes the following expression for the force of mortality

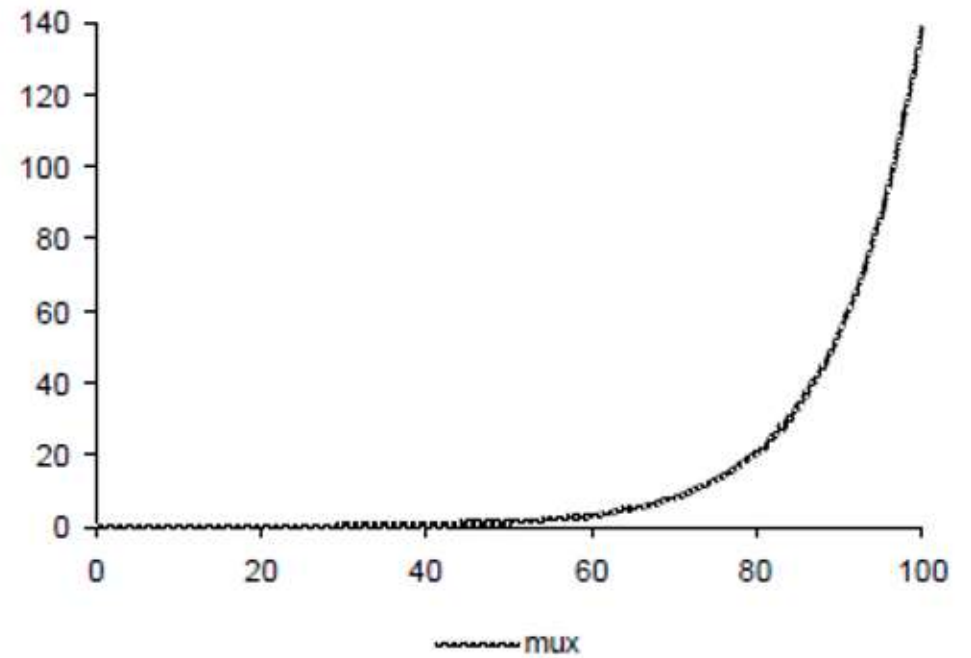
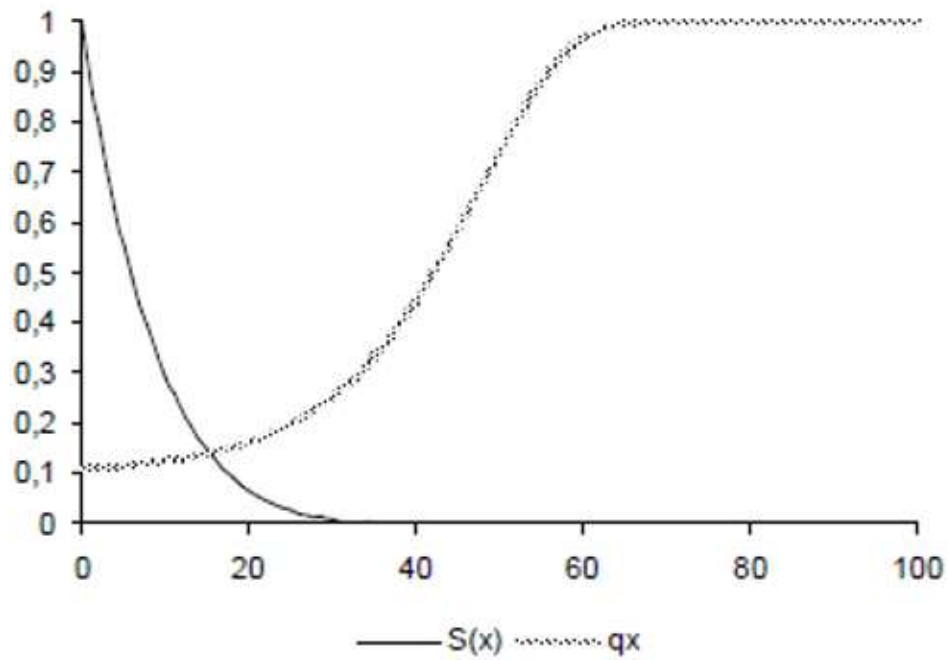
$$\mu_x = -\ln S - \ln g \cdot \ln C \cdot C^x$$

where $S < 1$, $g < 1$ and $C > 1$. Based on that assumption, we have that:

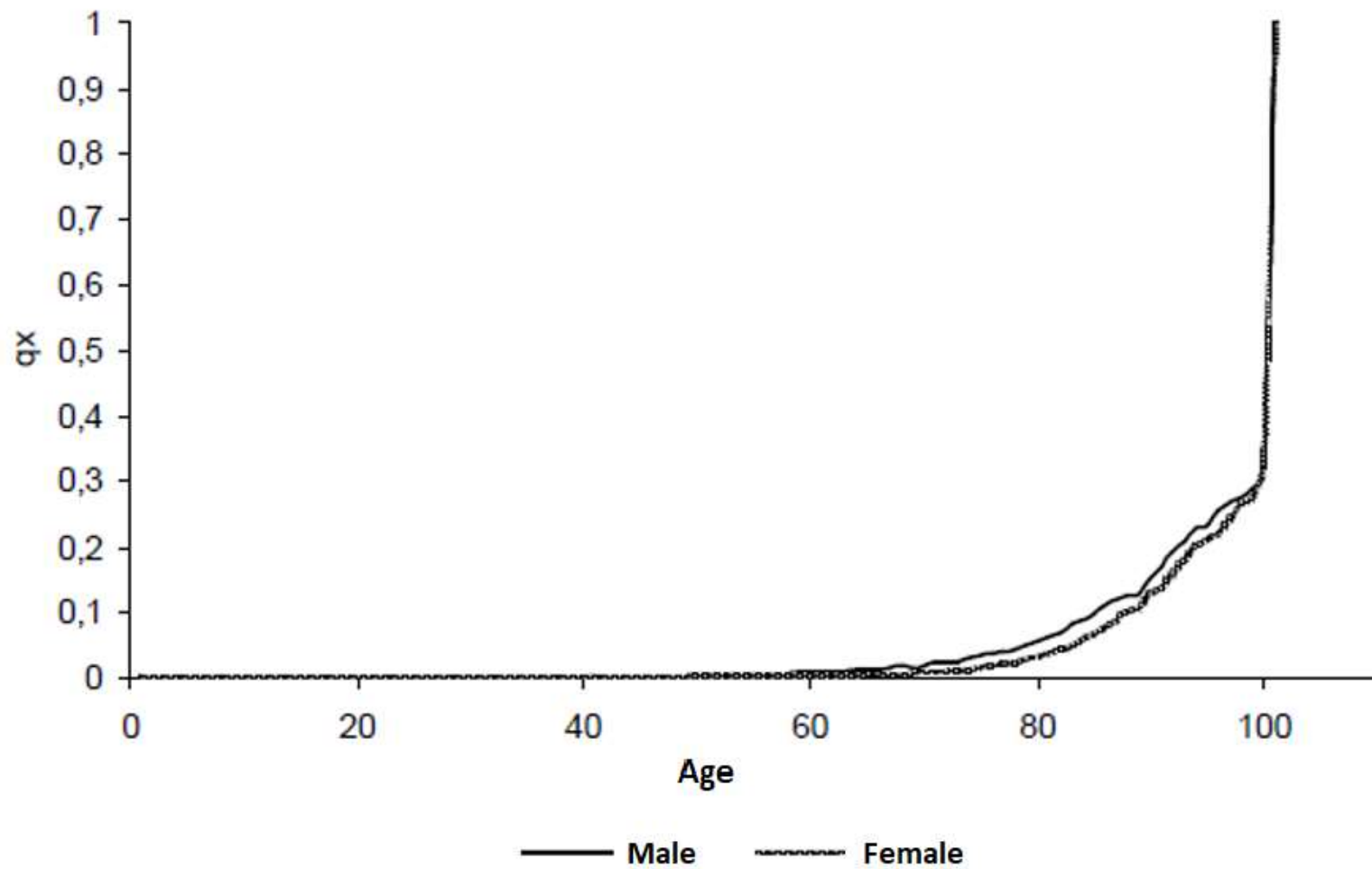
- $l_x = l_0 \cdot S^x \cdot g^{C^x-1}$
- ${}_np_x = S^n \cdot g^{C^x(C^n-1)}$; ${}_nq_x = 1 - S^n \cdot g^{C^x(C^n-1)}$
- $S(x) = S^x \cdot g^{C^x-1}$
- $f(x) = S^x \cdot g^{C^x-1} \cdot [-\ln S - \ln g \cdot \ln C \cdot C^x]$
- $g_x(t) = S^t \cdot g^{C^x(C^t-1)} \cdot [-\ln S - \ln g \cdot \ln C \cdot C^{x+t}]$
- $\bar{e}(x) = \int_0^\infty S^t \cdot g^{C^x(C^t-1)} dt$

It also exist a **Second Makeham Law**. If the first law is $\mu_x = A + BC^x$, the second law is $\mu_x = A + Hx + BC^x$

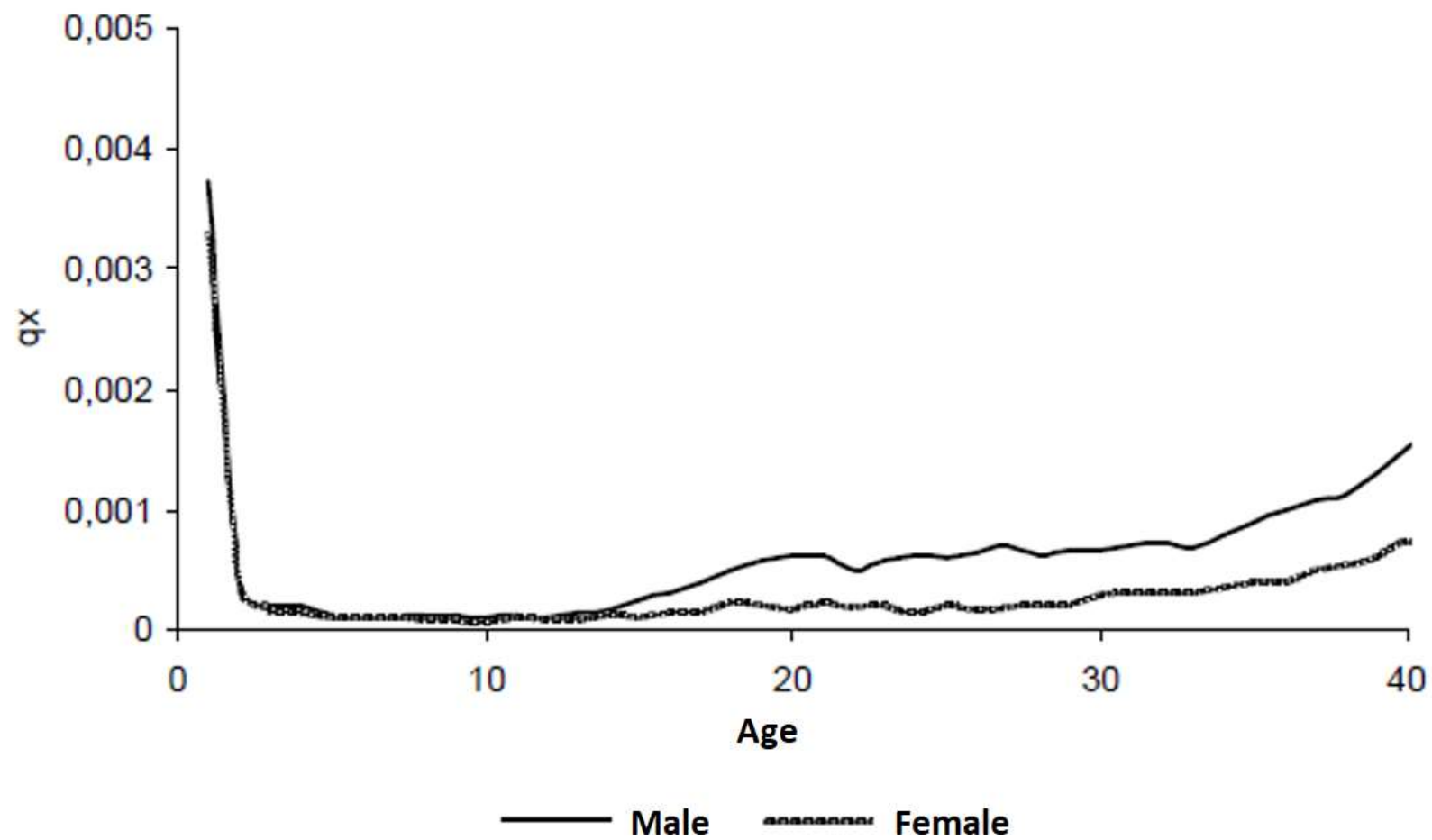
Makeham Law



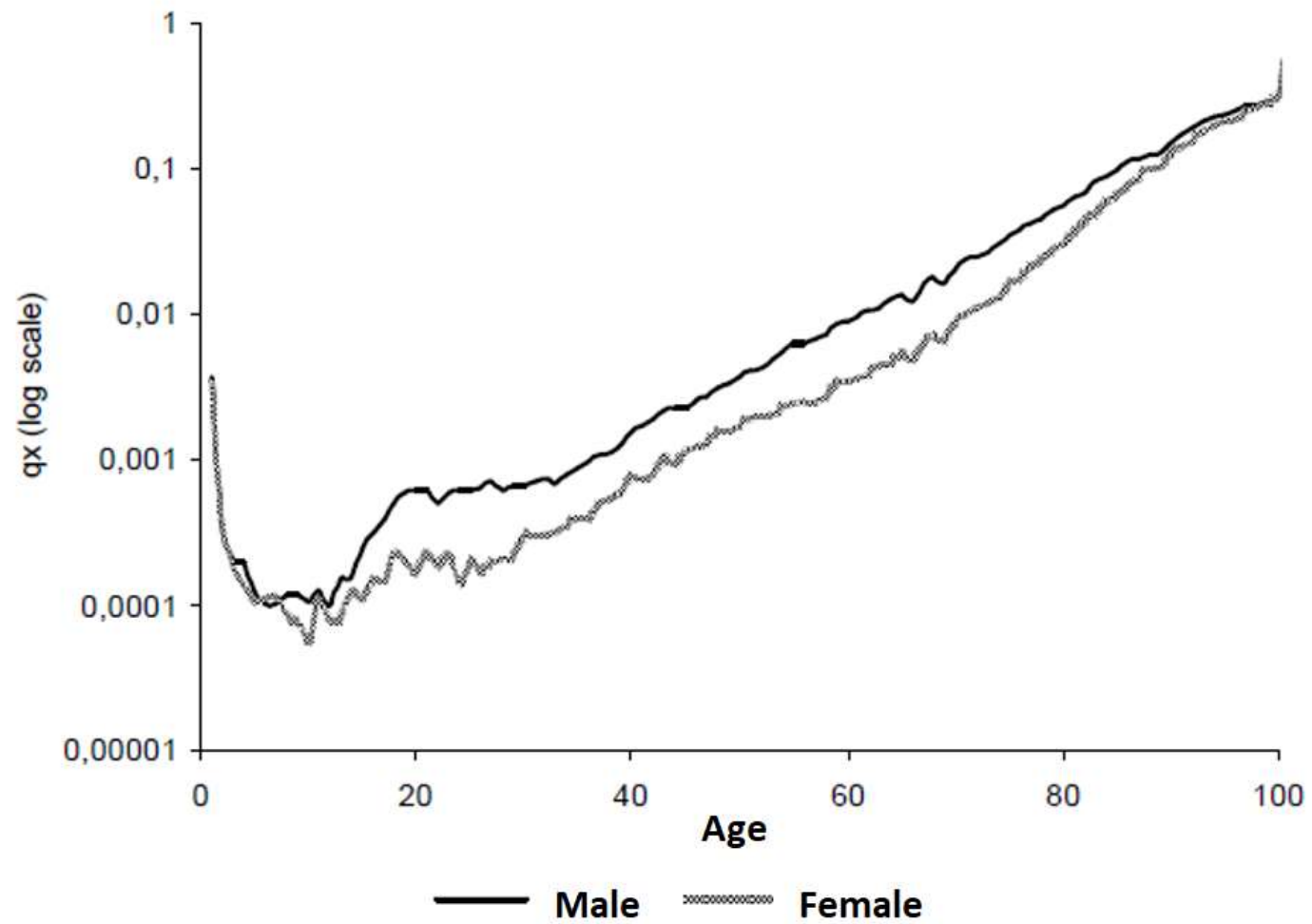
Plot of q_x



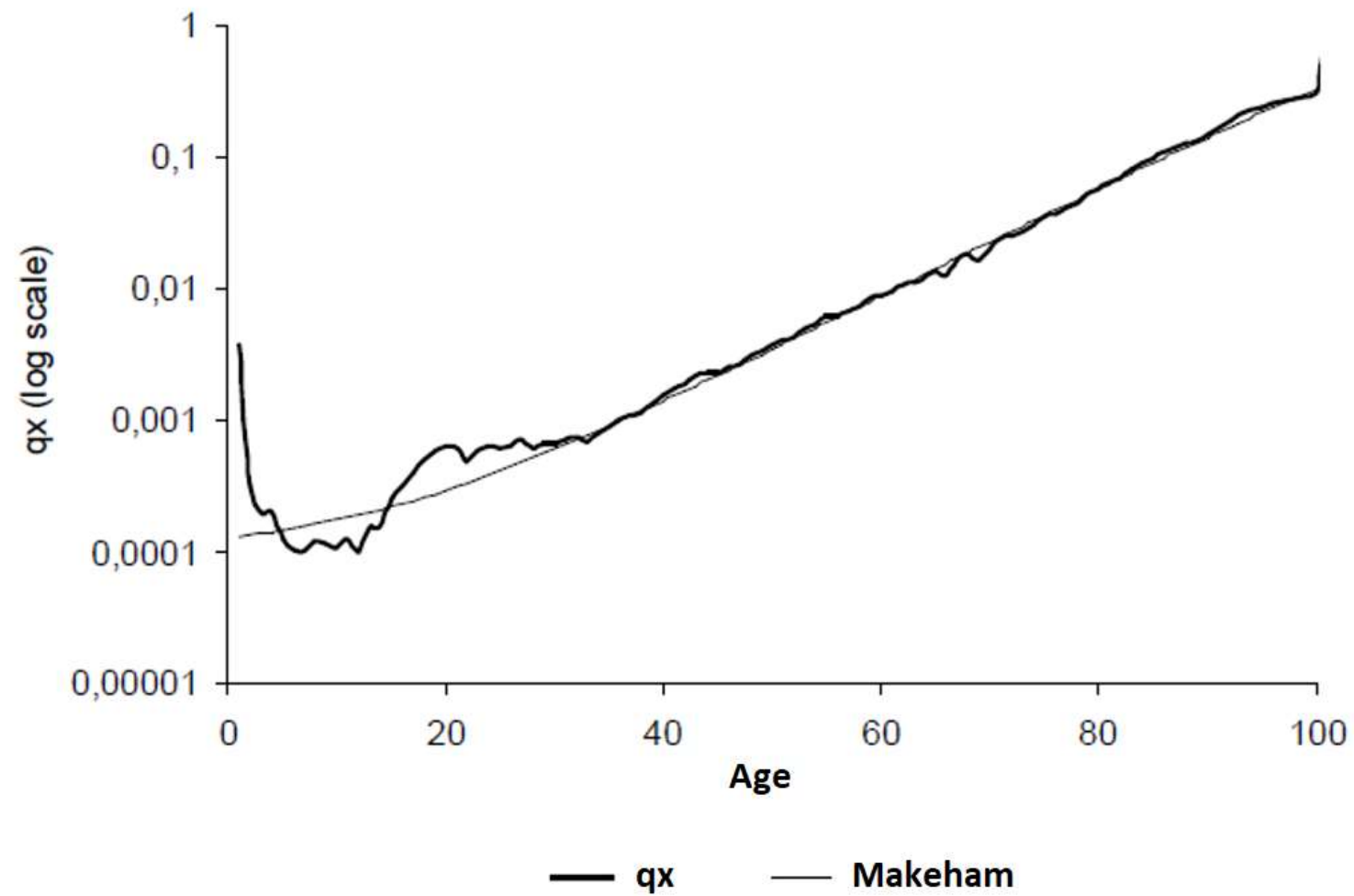
Plot of q_x



Plot of q_x



Plot of q_x



Heligman and Pollard Laws

- The **Heligman and Pollard law** (formulated in 1980) is a parametric model very popular and flexible. It has been widely used in actuarial science and demography.

- First law of Heligman and Pollard

$$q_x = A^{(x+B)^C} + De^{-E(\ln x - \ln F)^2} + \frac{GH^x}{1 + GH^x}$$

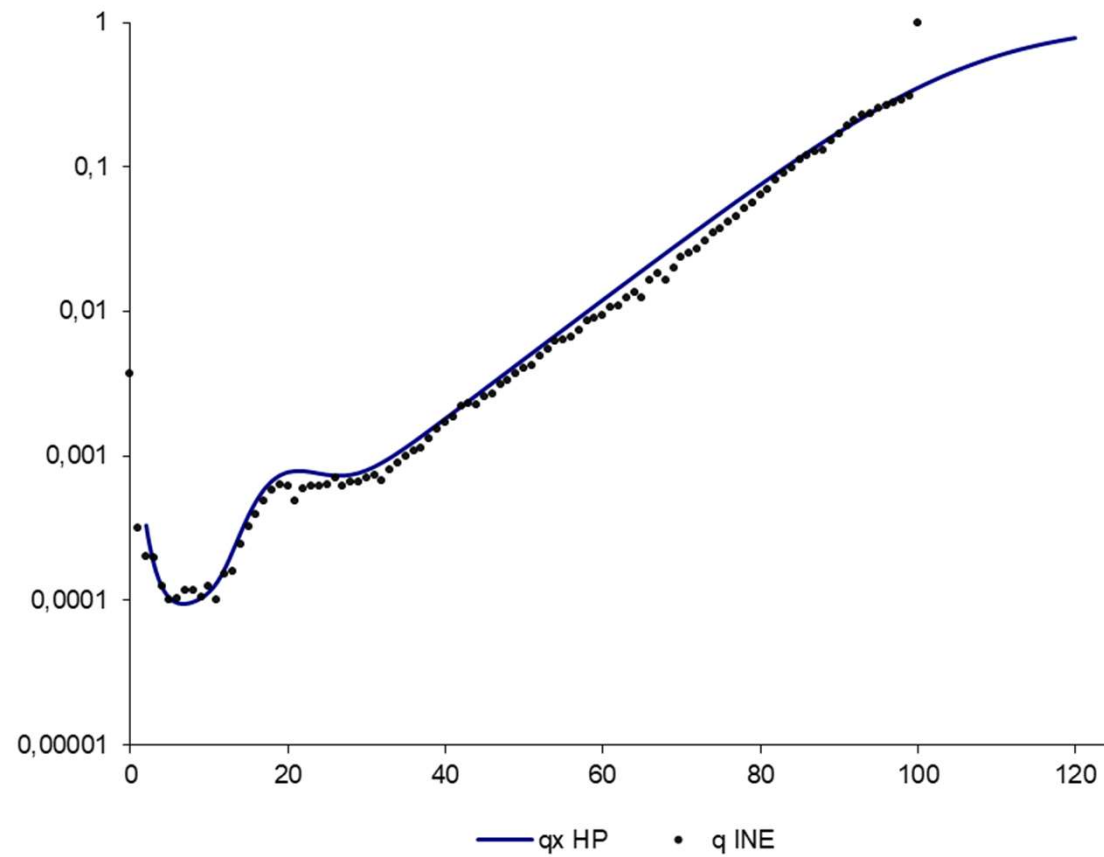
- Second law of Heligman and Pollard

$$q_x = A^{(x+B)^C} + De^{-E(\ln x - \ln F)^2} + \frac{GH^x}{1 + KGH^x}$$

- Third law of Heligman and Pollard

$$q_x = A^{(x+B)^C} + De^{-E(\ln x - \ln F)^2} + \frac{GH^{x^K}}{1 + GH^{x^K}}$$

Heligman and Pollard Laws



Kernel smoothing

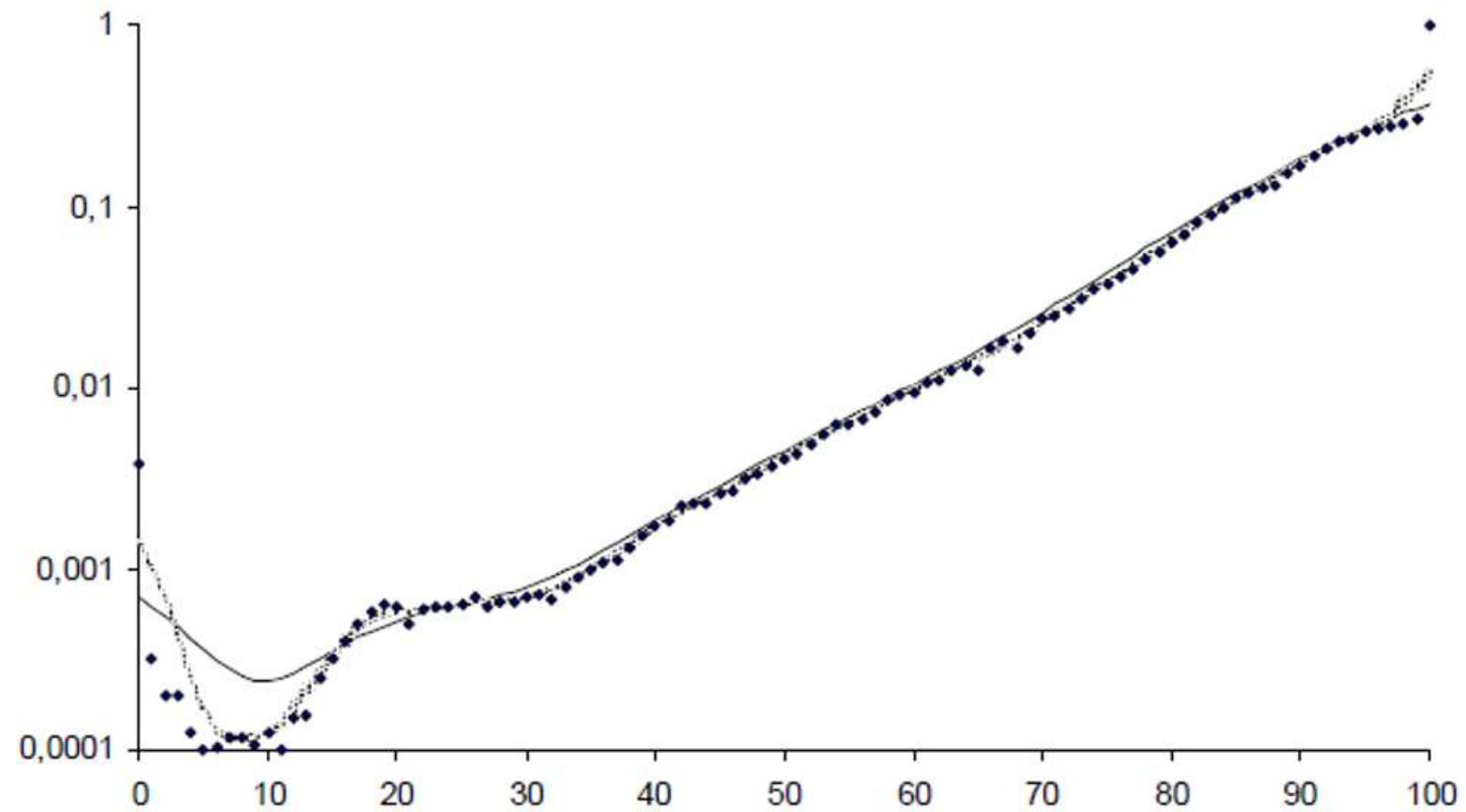
- Another possibility is to do a **kernel smoothing** of the raw mortality rates, which is a nonparametric approach, and therefore, it is not based on any parametric model. In that case,

$$\hat{q}_x = \frac{\sum_{z=1}^n q_z K_b(x - z)}{\sum_{z=1}^n K_b(x - z)}, \quad z = 1, \dots, n.$$

where $K_b(t) = b^{-1}K(t/b)$ and $K(\cdot)$ is the so-called kernel function (which, in order to obtain $\hat{q}_x$ gives a different weight to the values of q_z depending on how close they are to q_x) and b is the **smoothing parameter** (for larger b , the level of smoothing increases). A kernel function fulfils that $\sum_{-\infty}^{+\infty} K(t)dt = 1$. One of the most popular kernel functions is the one based on the standard normal distribution (also called Gaussian kernel)

$$K(t) = (2\pi)^{-1/2} \exp(-t^2/2), \quad \text{for } -\infty < t < +\infty$$

Kernel smoothing



Lee Carter Model

- The Lee Carter model (Lee and Carter, 1992) is expressed in terms of the central mortality rate. The central mortality rate is defined as $m_{x,t} = \frac{d_{x,t}}{E_{x,t}}$, where $d_{x,t}$ represents the number of deaths at age x in the period t with $t = t_1, \dots, t_n$ and $E_{x,t}$ is the censal function, that represents the number of individuals exposed to the risk of death.
- The Lee Carter model is expressed as follows,

$$\ln(m_{x,t}) = a_x + b_x^{(1)} k_t + \varepsilon_{x,t}$$

Lee Carter Model

$$\ln(m_{x,t}) = a_x + b_x^{(1)}k_t + \varepsilon_{x,t}$$

a_x : parameter (independent of the time) that captures the general pattern of mortality with respect to age.

$b_x^{(1)}$: parameter that represents the deviations of mortality for each age with respect to the global pattern.

k_t : parameter that captures the main temporal tendency (in log scale) in the mortality rates of all ages.

$\varepsilon_{x,t}$: error term, that represent all the aspects not captured by the model.

Lee Carter Model

In the original article, the authors (Lee and Carter, 1992) proposed to estimate the parameters by minimizing the mean squared error.

In order to guarantee that there is a unique solution Lee and Carter (1992) defined the following constraints:

$$\sum_x b_x^{(1)} = 1$$

$$\sum_t k_t = 0$$

Lee Carter Model

In the original article, the authors (Lee and Carter, 1992) proposed to estimate the parameters by minimizing the mean squared error, and using a two stage process. Specifically, the objective is to find the parameters that minimize:

$$\sum_{x,t} \left(\ln(m_{x,t}) - (a_x + b_x^{(1)} k_t) \right)^2$$

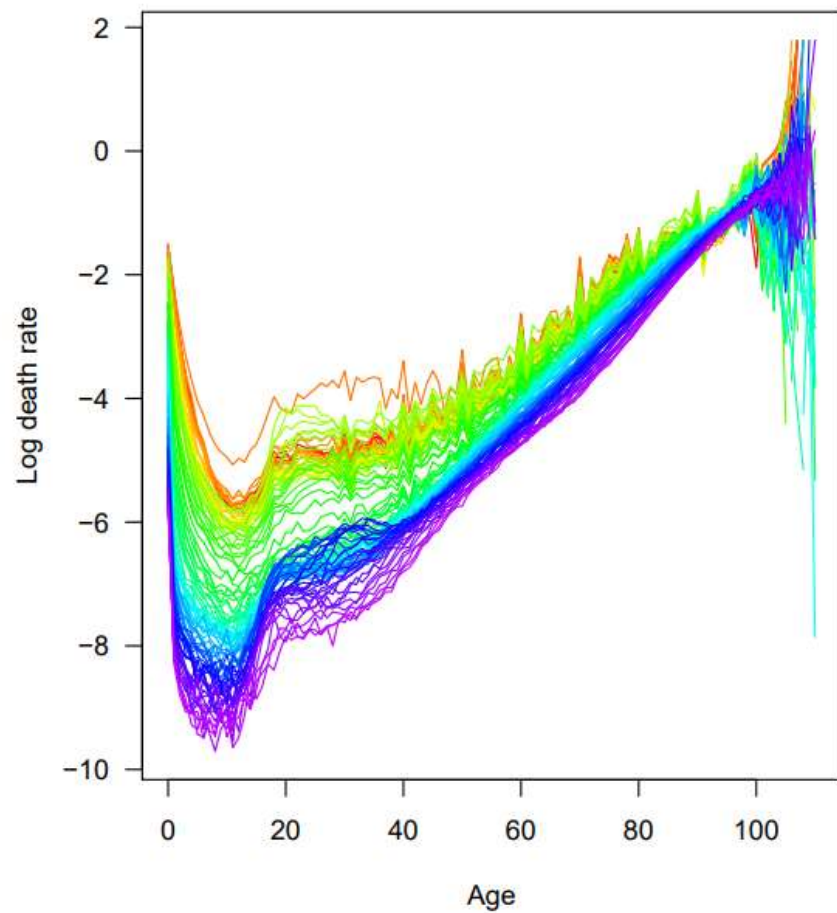
if we derive with respect to a_x and equal to 0, if we have into account the constraints, we have that:

$$\widehat{a}_x = \frac{1}{n} \sum_t \ln(m_{x,t})$$

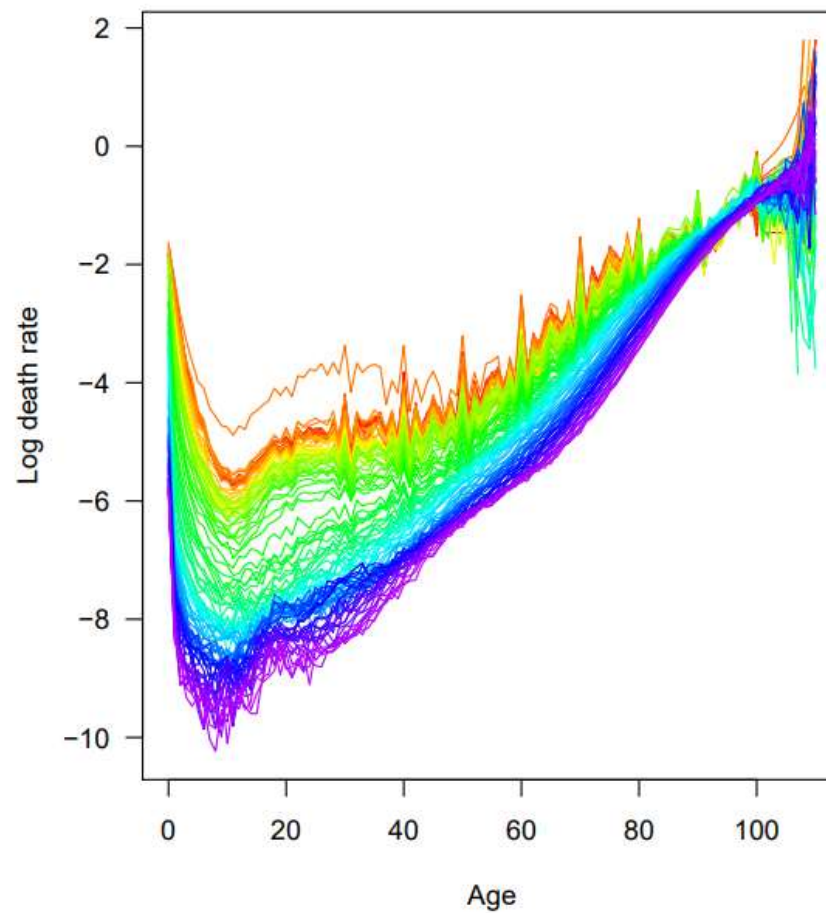
Then, the optimization algorithm performs the singular value decomposition of $Z_{x,t} = \ln(m_{x,t}) - \widehat{a}_x$, where $Z_{x,t} = b_x^{(1)} k_t$, and the estimation of $b_x^{(1)}$ and k_t is obtained. Then, mortality rates projections are obtained by performing time series analysis of k_t .

Lee Carter Model

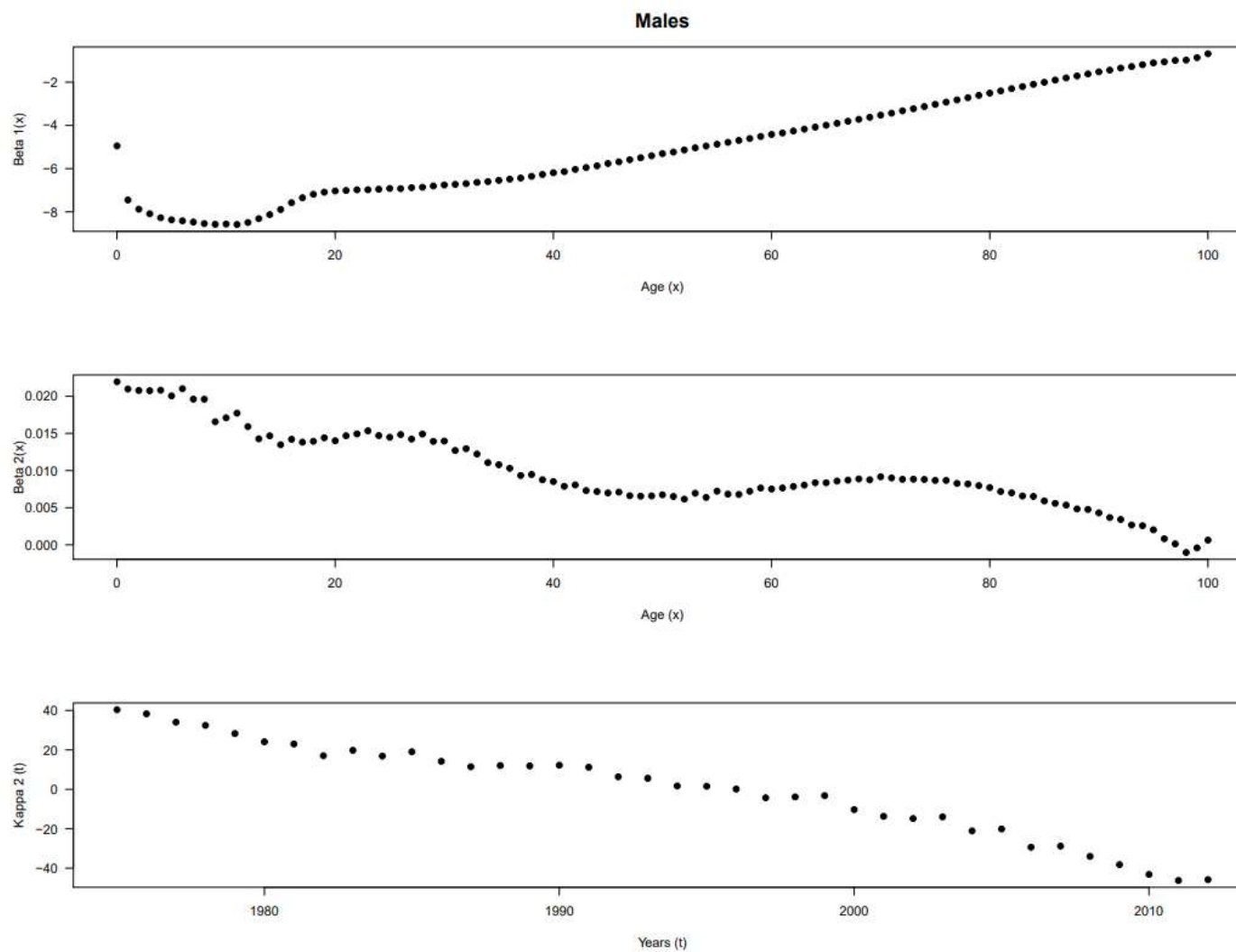
Spain: male death rates (1908–2012)



Spain: female death rates (1908–2012)

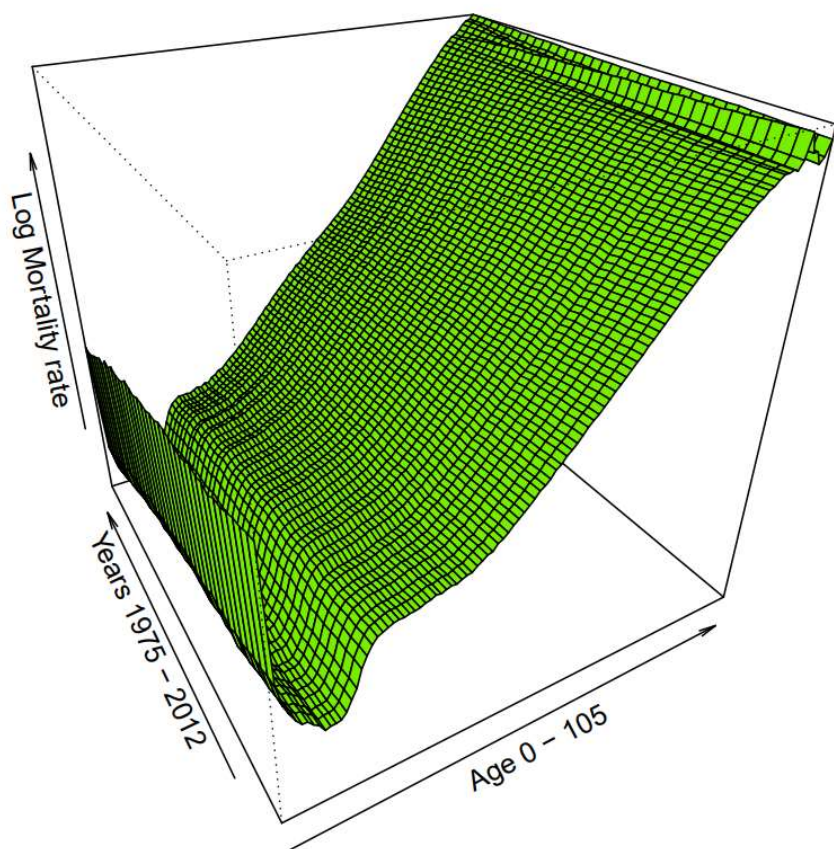


Lee Carter Model



Lee Carter Model

Log Mortality rates LC, Males 1975 – 2012



Spain: male death rates (2013–2042)

